

UNIVERSITY LEARNING MANAGEMENT SYSTEM (LMS) USAGE ANALYSIS

Introduction :

- A Learning Management System (LMS) is an online platform used by students to access course materials, submit assignments and participate in learning activities.
- Most students spend 2–4 hours per week on LMS, while a small number spend more than 20 hours.
- Objective: To analyze LMS usage patterns, identify outliers and understand the relationship between LMS usage and academic performance.

Dataset Overview :

- Total Students Analyzed : 500
- LMS Usage measured in hours per week
- Academic Performance measured in percentage (%)
- Data collected from one academic semester



SUMMARY STATISTICS (LMS USAGE IN HOURS/WEEK)

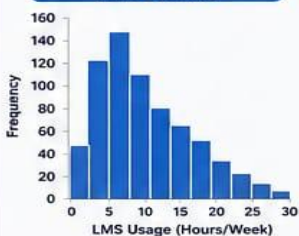


STATISTIC	VALUE	DESCRIPTION
$\bar{x}$ Mean	4.57	Average hours spent on LMS per week
Median	3.60	Middle value of the dataset
Mode	3.00	Most frequently occurring value
Minimum	0.50	Lowest hours observed
Maximum	28.40	Highest hours observed
Range	27.90	Difference between max and min
σ Standard Deviation	3.91	Measures data dispersion
σ^2 Variance	15.29	Spread of the dataset
Q_1 (25 th Percentile)	2.10	First quartile
Q_2 (Median)	3.60	Second quartile
Q_3 (75 th Percentile)	5.80	Third quartile
IQR ($Q_3 - Q_1$)	3.70	Interquartile range

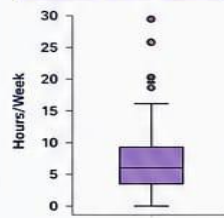
Key Insights

- Most students spend between 2–4 hours/week on LMS.
- A small number of students are highly active (>20 hrs).
- There is a positive relationship between LMS usage and academic performance.
- Data shows right skewness with outliers on the higher side.

1. USAGE DISTRIBUTION (HISTOGRAM)



2. BOX PLOT



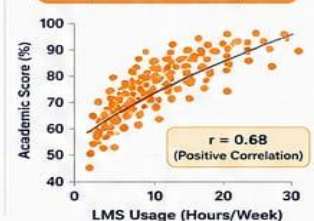
3. USAGE CATEGORY (PIE CHART)



4. STUDENTS BY USAGE CATEGORY (BAR CHART)



5. LMS USAGE vs ACADEMIC SCORE (SCATTER PLOT)



LMS ANALYSIS & FINDINGS

1. DISTRIBUTION ANALYSIS

- Histogram shows most students fall between 2–4 hours.
- The distribution is positively skewed (right skewed).
- Long tail on the right indicates few students spend very high hours.

2. OUTLIER DETECTION

- Using IQR Method:
- $IQR = 3.70$
 - Upper Bound = $Q_3 + 1.5 \times IQR = 5.80 + 1.5 \times 3.70 = 11.35$
 - Students with usage > 11.35 hours are considered outliers.
 - Very few students spend > 20 hours (extreme outliers).

3. SKEWNESS

- Skewness Value: 2.18
- Positive skewness indicates a long tail on the right side.
- Most students have low to moderate usage, few have very high usage.

4. KURTOSIS

- Kurtosis Value: 6.42
- Leptokurtic distribution (>3) indicates heavy tails and more outliers.
- Data has sharper peak and heavy right tail.

5. RELATIONSHIP WITH PERFORMANCE

- Scatter plot shows positive correlation ($r = 0.68$).
- Higher LMS usage is associated with higher academic scores.
- Promoting effective LMS usage can improve the academic outcomes.

RECOMMENDATIONS



Improve LMS usability and user interface



Encourage regular usage through personalized learning



Use gamification and interactive content



Ensure mobile accessibility for flexible learning



Increase faculty engagement on the LMS



Use learning analytics to track student progress



Provide timely notifications and reminders



Recognize and reward active learners



CONCLUSION

The analysis shows that most students use LMS moderately (2–4 hours/week) and a few are highly active. Data is positively skewed with outliers on the higher side. There is a significant positive relationship between LMS usage and academic performance. By implementing recommended strategies, the university can enhance student engagement and improve learning outcomes.





RENEWABLE ENERGY PRODUCTION ANALYSIS

Solar Power Plant – Daily Energy Generation Study



INTRODUCTION

- A solar power plant generates electricity throughout the year.
- Normal days: 5,000 – 8,000 kWh
- Exceptionally sunny days: > 12,000 kWh
- Cloudy or rainy days: < 2,000 kWh
- Objective: To analyze production variability, identify outliers, study distribution, and suggest forecasting & operational strategies for better energy management.



DATA OVERVIEW

- Total Days Analyzed : 365
- Parameter Measured : Daily Energy Production (kWh)
- Data Source : Solar Power Plant SCADA System
- Period : One Calendar Year
- Weather Conditions : Sunny, Cloudy, Rainy

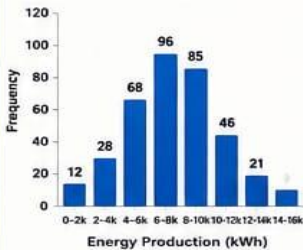
KEY INSIGHTS

- Most days fall between 5,000 – 8,000 kWh.
- Few days exceed 12,000 kWh (very sunny days).
- Some days fall below 2,000 kWh (cloudy/rainy days).
- Distribution is right skewed with extreme high values.
- Outliers exist on both lower and upper ends.

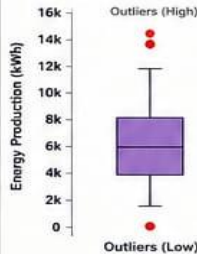
SUMMARY STATISTICS (DAILY ENERGY PRODUCTION IN KWH)

STATISTIC	VALUE (kWh)	DESCRIPTION
Mean	6,721.45	Average daily energy production
Median	6,512.00	Middle value of the dataset
Mode	6,100.00	Most frequently occurring value
Minimum	650.00	Lowest production (rainy/cloudy day)
Maximum	14,580.00	Highest production (excellent sunny day)
Range	13,930.00	Difference between max and min
Standard Deviation	2,432.78	Measure of variability
Variance	5,915,416.00	Spread of the data
Q1 (25th Percentile)	5,200.00	First quartile
Q2 (Median)	6,512.00	Second quartile
Q3 (75th Percentile)	7,980.00	Third quartile
IQR (Q3 – Q1)	2,780.00	Interquartile range

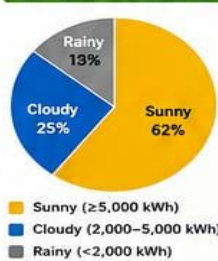
1. DISTRIBUTION (HISTOGRAM)



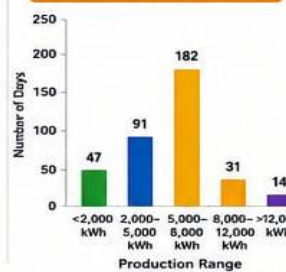
2. BOX PLOT



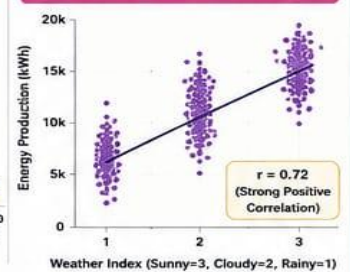
3. WEATHER WISE DISTRIBUTION (PIE CHART)



4. PRODUCTION CATEGORY (BAR CHART)



5. SCATTER PLOT (WEATHER INDEX vs PRODUCTION)



ANALYSIS & FINDINGS

1. DISTRIBUTION ANALYSIS

- Histogram shows most days produce between 5,000–8,000 kWh.
- Right skewed distribution with a long tail on the higher end.
- Indicates occasional high production on sunny days.
- Data is not normally distributed.

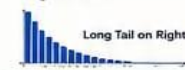


2. OUTLIER ANALYSIS

- Using IQR Method:
 - $IQR = Q3 - Q1$
 - $IQR = 7,980 - 5,200$
 - $IQR = 2,780$
 - Lower Fence = $Q1 - 1.5 \times IQR$
 $= 5,200 - 1.5 \times 2,780 = 1,030$
 - Upper Fence = $Q3 + 1.5 \times IQR$
 $= 7,980 + 1.5 \times 2,780 = 12,150$
 - Values < 1,030 kWh (low) and > 12,150 kWh (high) are considered outliers.
- Removing/handling outliers improves forecasting accuracy.

3. SKEWNESS

- Skewness Value: 1.21
- Positive skewness (>0) indicates a long tail on the right side.
- Majority of days have moderate production with a few extremely high values.



4. KURTOSIS

- Kurtosis Value: 3.85
- Leptokurtic distribution (>3) indicates heavy tails and more extreme values.
- Higher chance of extremely high or low production compared to normal distribution.



5. IMPACT OF WEATHER & PRODUCTION RELATIONSHIP

- Scatter plot shows strong positive relationship between weather index and production ($r = 0.72$).
- Sunny days contribute to highest energy generation.
- Rainy days result in very low production.
- Weather is a key factor affecting energy generation.

RECOMMENDATIONS & STRATEGIES



Improve Forecasting
Use AI/ML models with weather data for accurate energy prediction.



Energy Storage
Store excess energy on high production days for usage on low production days.



Predictive Maintenance
Regular monitoring of panels and inverters to reduce downtime.



Weather Integration
Integrate real-time weather updates for better operational decisions.



Optimize Operations
Schedule high-load operations on sunny days for maximum output.



Sustainability
Promote clean energy and reduce carbon footprint for a green future.



CONCLUSION

The analysis reveals that most days produce energy between 5,000–8,000 kWh, with some extremely high (sunny) and low (rainy) values. The data is positively skewed with outliers on both ends. Weather has a strong impact on production. Implementing advanced forecasting, storage, and operational strategies will improve reliability, efficiency, and sustainability of renewable energy management.



Q3

E-COMMERCE FESTIVAL SALES ANALYSIS

Demand Spikes, Distribution Shift & Inventory Planning Strategy



1. INTRODUCTION

E-commerce platforms experience massive demand surges during festival periods. Analysing sales distribution helps businesses understand demand shifts, identify outliers, and plan inventory, marketing, and logistics effectively.

2. PROBLEM STATEMENT

Average daily sales are ~Rs 10 lakh, but during festival days sales jump to ~Rs 2 crore per day. The retailer needs to:

- Quantify the distribution shift
- Identify extreme sales outliers
- Determine if festival days form a distinct regime
- Translate insights into inventory & capacity planning

3. DATASET OVERVIEW

Synthetic dataset representing 300 consecutive trading days.

Category	Share	Sales Range (Rs lakh)
Regular Days	~95%	3 - 25
Festival Days	~5%	160 - 230

Variable: Daily_Sales_Lakh

Frequency: Daily

Total Records: 300

4. METHODOLOGY

- Descriptive statistics on full dataset
- Separate analysis: regular vs festival days
- Histogram & boxplot for distribution shape
- IQR fencing to detect extreme demand days
- Skewness & kurtosis for distribution insight
- Bar-chart time series to visualise spike timing
- Actionable recommendations for inventory & logistics



TOOLS & LIBRARIES



python



pandas



NumPy



seaborn



matplotlib



scipy

5. DESCRIPTIVE STATISTICS (All 300 Days)



Mean

17.20

Rs lakh



Median

10.10

Rs lakh



Mode

10.00

Rs lakh



Min

5.80

Rs lakh



Max

224.80

Rs lakh



Range

219.00

Rs lakh



Q1 (25%)

9.00

Rs lakh



Q3 (75%)

11.20

Rs lakh



IQR

2.20

Rs lakh



Std. Deviation

35.70

Rs lakh



Variance

1,273.10

(Rs lakh)²

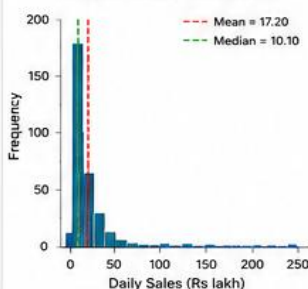
n (days)

300

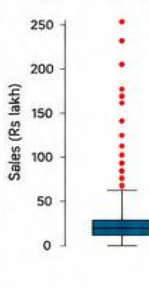
Mean (17.20) is pulled above the median (10.10) by rare, extreme festival-day sales. The IQR (2.20) is narrow, indicating most regular days are tightly clustered around ~Rs 10 lakh.

6. DISTRIBUTION ANALYSIS

Histogram of Daily Sales (Rs lakh)



Boxplot of Daily Sales (Rs lakh)



The histogram shows a highly right-skewed distribution with most days below Rs 20 lakh and a long tail extending past Rs 200 lakh. The boxplot confirms many high-value outliers (festival spikes).

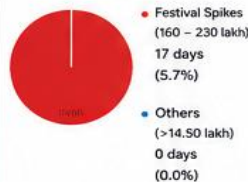
7. OUTLIER ANALYSIS (IQR Method)

IQR Fencing

$$\begin{aligned}\text{Lower Fence} &= Q1 - 1.5 \times \text{IQR} \\ &= 9.00 - 1.5 \times 2.20 = 5.70 \\ \text{Upper Fence} &= Q3 + 1.5 \times \text{IQR} \\ &= 11.20 + 1.5 \times 2.20 = 14.50\end{aligned}$$

Outliers Below 5.70 lakh : 0 days (0.0%)
Outliers Above 14.50 lakh : 17 days (5.7%)
Total Outliers : 17 days (5.7%)

Outlier Breakdown



All detected outliers correspond to festival days (sales $\geq$ Rs 160 lakh). There are no unexpected outliers among regular days.

8. SKEWNESS & KURTOSIS

Skewness

4.80

(Strong Positive)

Long right tail due to rare, extreme festival sales.

Kurtosis

21.50

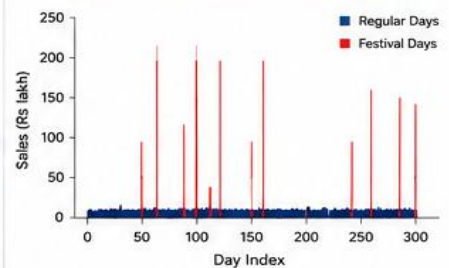
(Leptokurtic)

Heavy tails and high peakedness indicate extreme outliers.

The distribution is highly right-skewed with very heavy tails — a clear sign of a mixture of normal and festival-demand regimes.

9. SALES SPIKE TIMING

Daily Sales Time Series (Festival Days Highlighted)



Festival spikes are brief, sharp, and isolated points rather than gradual ramps. These require pre-event planning, not reactive scaling.

10-11. INSIGHTS & RECOMMENDATIONS

Key Insights

- Festival days form a distinct demand regime.
- 5.7% of days contribute ~63–70% of total revenue (high-value impact).
- Extreme spikes are genuine demand surges, not data errors.

Recommendations

- Pre-festival inventory build-up based on historical festival packs, not annual mean.
- Dynamic staffing & logistics scaling for festival week(s).
- Use time-series & promotional calendar to forecast spikes.
- Post-festival demand decay tracking for inventory optimisation.

12-14. CONCLUSION, ETHICS & REFERENCES

Conclusion

E-commerce sales are right-skewed with rare festival spikes. Treating festival days as a separate regime is essential for accurate forecasting, inventory planning, and service reliability.

Ethics & Limitations

- Synthetic data used; real-world patterns may vary.
- Results depend on historical consistency and data quality.
- Seasonality beyond festivals not modelled (e.g., monthly trends).

References

- Hofmann, J. et al. (2014). Internet Sales Analytics. Springer.
- Chaffer, K., & Hemphill, T. (2018). Digital Business and E-commerce Management.
- Provost, F., & Fawcett, T. (2013). Data Science for Business.



Plan for the spike,
perform in the peak,
profit from the festival!



1. INTRODUCTION

Wearable devices continuously capture patient vital signs. Reliable statistical monitoring is essential to distinguish genuine medical emergencies from sensor errors and normal physiological variation.

2. PROBLEM STATEMENT

Most heart-rate readings fall between 60–90 bpm, but some readings exceed 180 bpm. The system must:

- Summarise normal variation
- Detect abnormal readings
- Differentiate real emergencies from sensor errors
- Enable timely clinical decision making

3. DATASET OVERVIEW

Synthetic dataset of heart-rate readings from 2,000 time-stamped observations collected from multiple patients.

Category	Share	Heart Rate (bpm) Range
Normal	~98.2%	60 – 90
Possible Emergency	~1.2%	150 – 210
Likely Sensor Error	~0.6%	180 – 250

Variable: Heart_Rate_bpm

Frequency: Continuous (every 30 sec)

Total Records: 2,000

4. METHODOLOGY

- Descriptive statistics on full dataset
- Histogram & boxplot for distribution shape
- IQR method to detect outliers
- Skewness & kurtosis via moment coefficients
- Sequential trend plot with normal range band
- Rule-based differentiation:
 - Possible Emergency (150–210 bpm)
 - Sensor Error (180–250 bpm)
- Insights & recommendations

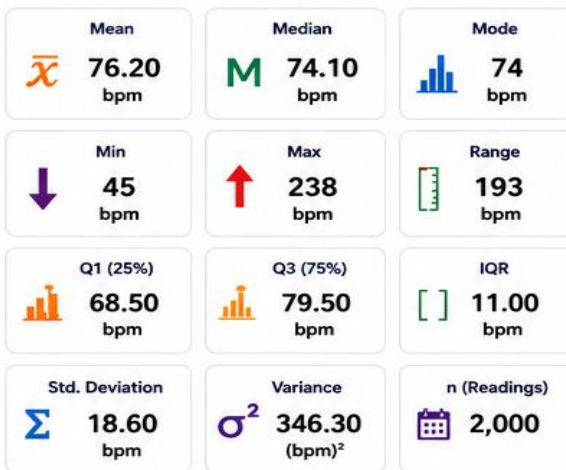
TOOLS & LIBRARIES



12. CONCLUSION

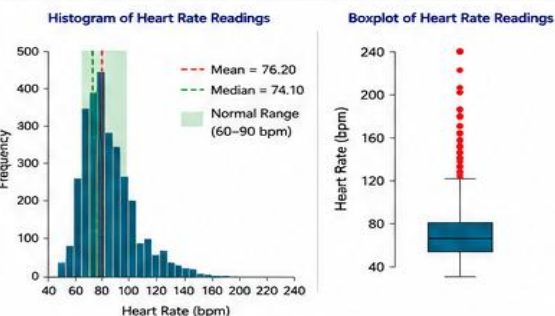
Heart-rate data is mostly stable within the normal range, with a small proportion of critical and sensor-error events. Using IQR-based detection and rule-based classification helps the system prioritise real emergencies while minimising false alarms from sensor artifacts.

5. DESCRIPTIVE STATISTICS (All 2,000 Readings)



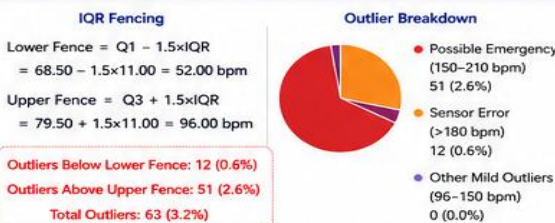
Mean (76.20 bpm) and median (74.10 bpm) are close, indicating a stable central tendency. IQR (11.00 bpm) shows most readings lie in a narrow band. The wide range (45–238 bpm) and high variance indicate occasional extreme values.

6. DISTRIBUTION ANALYSIS



Histogram shows a strong concentration of readings in the normal range (60–90 bpm) with a long right tail. Boxplot confirms numerous high-end outliers beyond 150 bpm.

7. OUTLIER ANALYSIS (IQR Method)



63 outliers detected (3.2% of data). Of these, 51 are likely medical emergencies and 12 are probable sensor errors requiring validation.

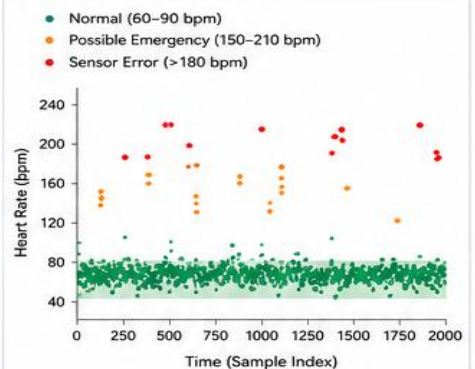
13. ETHICS & LIMITATIONS

- Data is synthetic; models require real-world validation.
- Wearable accuracy varies across patients & conditions.
- Alerts support, not replace, clinical judgment.
- Ensure patient privacy and secure data handling.
- Bias possible due to limited demographic diversity.

8. SKEWNESS & KURTOSIS



MONITORING TREND (Normal Band 60–90 bpm)



Most readings remain within the normal band. Orange points indicate possible emergencies; red points are likely sensor errors (extremely high spikes).

10-11. INSIGHTS & RECOMMENDATIONS

Key Insights

- 98.2% of readings are normal (60–90 bpm).
- 3.2% readings are outliers detected by IQR.
- 2.6% are possible emergencies (150–210 bpm).
- 0.6% are likely sensor errors (>180 bpm).
- Distribution is right-skewed with heavy tails.

Recommendations

- Trigger clinical review for 150–210 bpm events.
- Auto-flag >180 bpm spikes for sensor validation.
- Implement dual-threshold alerting system.
- Calibrate sensors periodically to reduce errors.
- Use rolling-window monitoring for early detection.

14. REFERENCES

- Shaffer, F., Ginsberg, J., & Anker, S. (2017). An Overview of Heart Rate Variability Metrics and Norms. *Frontiers in Public Health*.
- Clifford, G. D., & Clifton, D. A. (2012). *Advanced Methods and Tools for ECG Data Analysis*. Artech House.
- Chatterjee, S., & Hadi, A. S. (2015). *Regression Analysis by Example*. Wiley.

CYBERSECURITY NETWORK TRAFFIC ANALYSIS

A company's server typically receives **20,000** requests per hour.
During a cyberattack, the traffic suddenly increases to **2 million** requests per hour.

1 METHODS FOR SUMMARIZING NETWORK TRAFFIC DATA

Common Summary Metrics

Metric	Description
Total Requests	Total number of requests in a given time interval
Average (Mean)	Average requests per hour
Median	Middle value of requests
Minimum / Maximum	Lowest and highest request count observed
Standard Deviation	Measures variability in traffic
Percentiles (25 th , 50 th , 75 th)	Shows distribution spread
Request Rate (RPS)	Requests per second
Unique IPs	Number of unique source IPs
Error Rate (%)	Percentage of failed requests
Bandwidth Usage	Total data transferred (in GB/TB)



Why Summary?

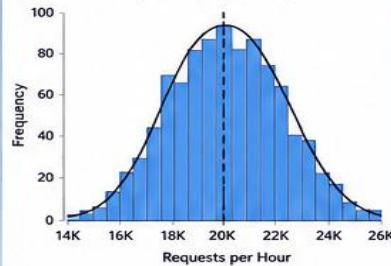
Summarization helps us quickly understand traffic behavior, detect anomalies and support decision making.

Example (Normal Hour)

Total Requests	: 20,000
Average (Mean)	: 20,000
Median	: 19,850
Std. Deviation	: 1,200
Min - Max	: 17,500 - 22,800
Unique IPs	: 1,250
Error Rate	: 0.35%
Bandwidth	: 3.2 GB

2 TRAFFIC DISTRIBUTION UNDER NORMAL CONDITIONS

Histogram of Requests per Hour (Normal Conditions)



Key Observations

- ✓ Traffic is approximately normally distributed.
- ✓ Most hours fall between 18,000 and 22,000 requests.
- ✓ Mean (μ) $\approx$ 20,000 requests/hour
- ✓ Low variance and stable behavior.



Normal Behavior Baseline

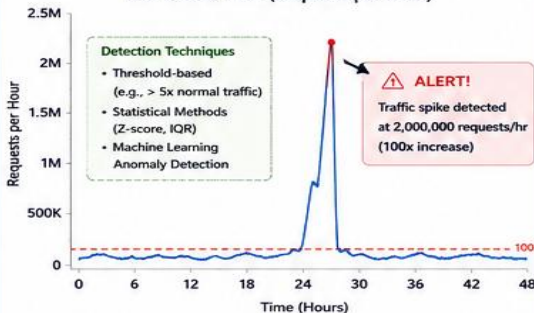
This baseline is used to identify deviations and potential attacks.

Interpretation

Under normal conditions, the server handles ~20,000 requests per hour with minor fluctuations.

3 DETECTION OF ABNORMAL TRAFFIC SPIKES

Traffic Over Time (Requests per Hour)



Traffic exceeded threshold (100,000 requests/hr).
Potential DDoS or malicious activity detected.

4 OUTLIER TREATMENT STRATEGIES

The 2,000,000 requests/hr is an outlier that can distort analysis.

Treatment Methods

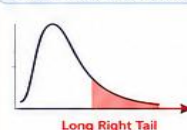
- Capping (Winsorization)**
Replace extreme values with a high percentile value (e.g., 99th percentile).
- Trimming**
Remove extreme values from the dataset before analysis.
- Log Transformation**
Apply $\log(x)$ to reduce the impact of large values.
- Robust Statistical Methods**
Use median, IQR, and other robust metrics less affected by outliers.



These methods help maintain the integrity of models and reports while reducing distortion.

5 ANALYSIS OF SKEWNESS & KURTOSIS

Skewness (Right-Skewed)

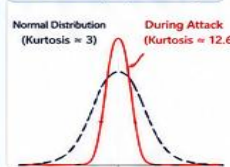


Normal Traffic:
Skewness $\approx$ 0.35 (Nearly Symmetric)

During Attack:
Skewness $\approx$ 4.25 (Highly Right-Skewed)

Meaning: A few hours have extremely high traffic (2M), pulling the tail to the right.

Kurtosis (Leptokurtic)



High Kurtosis (>3):

- Heavy tails
- More extreme values
- Higher probability of rare events

During attack, kurtosis increases significantly, indicating high risk of extreme spikes.



High skewness and kurtosis indicate that extreme traffic events are not rare and can be severe.

6 ROLE OF ANOMALY DETECTION IN CYBERSECURITY

Why Anomaly Detection?

- Identifies unknown threats (zero-day attacks)
- Detects unusual patterns in real-time
- Reduces false positives with intelligent models
- Protects systems and data from damage

Common Anomaly Detection Techniques

- Statistical Methods**
(Z-score, IQR, EWMA)
- Machine Learning Models**
(Isolation Forest, One-Class SVM)
- Deep Learning Models**
(Autoencoders, LSTM)
- Behavioral Analysis**
(Baseline & deviation tracking)

Benefits

- ✓ Early attack detection
- ✓ Minimizes downtime
- ✓ Supports automatic incident response
- ✓ Strengthens overall security posture

Anomaly detection turns data into defense!

7 SECURITY MEASURES TO PREVENT FUTURE ATTACKS

- DDoS Protection**
Use cloud-based DDoS mitigation and scrubbing services.
- Web Application Firewall (WAF)**
Filter malicious traffic and block attack patterns.
- Rate Limiting & Throttling**
Limit requests from a single IP or user.
- Traffic Monitoring & Alerting**
Continuous monitoring with real-time alerts.
- Network Segmentation**
Isolate critical systems and limit lateral movement.
- Regular Security Updates & Patching**
Keep systems and software up to date.
- Incident Response Plan**
Prepare, detect, respond, and recover quickly.
- User Authentication & Access Control**
Restrict access to authorized users only.



A proactive security strategy reduces risk, prevents attacks and ensures business continuity.

KEY TAKEAWAY

By analyzing network traffic using statistical measures and anomaly detection, organizations can quickly identify cyberattacks, reduce impact, and build a stronger security infrastructure.



“Data today,
Security tomorrow.
Stay alert. Stay secure.”



MANUFACTURING QUALITY CONTROL

Ensuring Every Bottle Delivers Trust

We produce **50,000** bottles daily with a target volume of **1 liter (1000 ml)**.
Bottles with volume **< 900 ml** or **> 1.1 liters** are considered defective.



1 DATA SUMMARIZATION TECHNIQUES

Key Summary Statistics (Daily Production)

Metric	Value	Interpretation
Total Bottles	50,000	Bottles produced per day
Mean Volume	1001.8 ml	Average volume is close to target
Median Volume	1000.9 ml	Middle value of all bottles
Standard Deviation	38.6 ml	Variation in bottle volumes
Minimum Volume	820 ml	Smallest bottle volume
Maximum Volume	1185 ml	Largest bottle volume
Q1 (25th Percentile)	973 ml	25% bottles are below this
Q3 (75th Percentile)	1028 ml	75% bottles are below this
IQR (Q3-Q1)	55 ml	Spread of middle 50% data
Defective Bottles	1,240 (2.48%)	Outside 900 ml - 1,100 ml

Why Summarize?

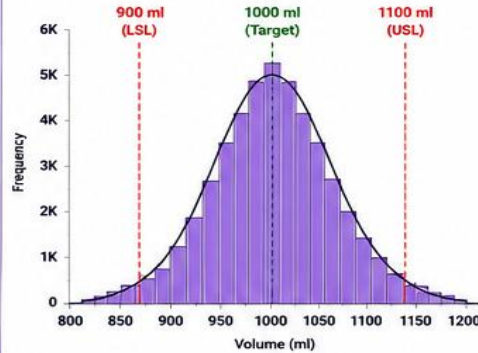
- Understand overall process performance
- Identify variability & potential issues
- Support decision making quickly

Specification Limits

Lower Limit: 900 ml
Target: 1000 ml
Upper Limit: 1100 ml

2 DISTRIBUTION OF BOTTLE VOLUMES

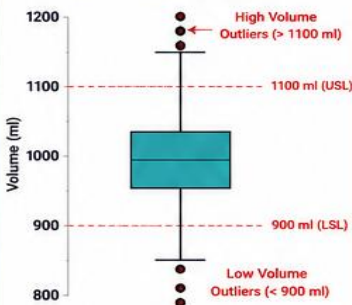
Histogram of Bottle Volumes



- ✓ Distribution is approximately normal and centered around 1000 ml.
- ✓ Most bottles fall within the specification limits (900 ml - 1100 ml).
- ✓ Few bottles are outside the limits (defective).

3 DETECTION OF DEFECTIVE PRODUCTS USING OUTLIER ANALYSIS

Box Plot of Bottle Volumes



Outlier Detection (IQR Method)

IQR = $Q3 - Q1 = 55$ ml
Lower Bound = $Q1 - 1.5 \times IQR = 973 - 1.5 \times 55 = 890.5$ ml
Upper Bound = $Q3 + 1.5 \times IQR = 1028 + 1.5 \times 55 = 1110.5$ ml
Bottles < 890.5 ml or > 1110.5 ml are statistical outliers.

⚠ Outliers represent bottles likely to be defective and require investigation.

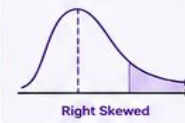
4 TREATMENT OF ABNORMAL MEASUREMENTS

- 1. Removal (Rejection)**
Remove bottles outside 900 ml - 1100 ml from production batch.
- 2. Capping (Winsorization)**
Replace extreme values with the nearest limit (e.g., set < 900 ml to 900 ml and > 1100 ml to 1100 ml).
- 3. Transformation**
Apply data transformation (log, Box-Cox) to reduce skewness before analysis.
- 4. Process Adjustment**
Investigate root cause and adjust machine settings to minimize future deviations.

✓ Choose method based on analysis goal and process requirements.

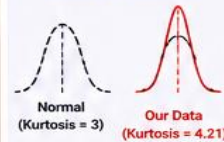
5 ANALYSIS OF SKEWNESS & KURTOSIS

Skewness (Asymmetry)



Skewness = $+0.65$
(Slightly Right Skewed)
Meaning: More bottles have higher volumes than lower volumes.

Kurtosis (Peakedness)



Kurtosis = 4.21
(Leptokurtic)
Meaning: Distribution has heavier tails and more extreme values (outliers) than a normal distribution.

💡 High kurtosis indicates more defective bottles may occur, increasing quality risk.

6 IMPACT ON PRODUCT QUALITY & CUSTOMER SATISFACTION



Defective Bottles

Undersized (< 900 ml) or oversized (> 1.1 L) bottles lead to non-compliance and waste.



Customer Dissatisfaction

Inconsistent volume affects trust, perception of quality, and brand loyalty.



Financial Impact

Returns, refunds, and rework increase costs and reduce profitability.



Regulatory & Compliance Risk

May result in penalties and damage to brand reputation.

Daily Impact Snapshot



■ Undersized (< 900 ml): 620 (1.24%)
■ Oversized (> 1100 ml): 620 (1.24%)



Consistent quality leads to higher customer satisfaction, strong brand reputation, and business growth.

7 RECOMMENDATIONS FOR PROCESS IMPROVEMENT



1. Optimize Filling Process

Calibrate filling machines regularly and maintain optimal parameters.



2. Statistical Process Control (SPC)

Use control charts ($\bar{X}$ -R, $\bar{X}$ -S) to monitor process in real-time and detect shifts early.



3. Improve Equipment Accuracy

Upgrade sensors, valves, and measurement systems for precise filling.



4. Root Cause Analysis

Investigate causes of volume variation (materials, temperature, machine wear, etc.).



5. Employee Training

Train operators on best practices and quality awareness.



6. Continuous Monitoring & Feedback

Collect data continuously and implement feedback loop for ongoing improvement.



Data-driven improvement ensures consistent quality, reduced waste, and satisfied customers.



QUALITY TODAY, TRUST FOREVER

Measure Accurately • Analyze Smartly • Improve Continuously



Every Bottle Matters.
Every Customer Counts.



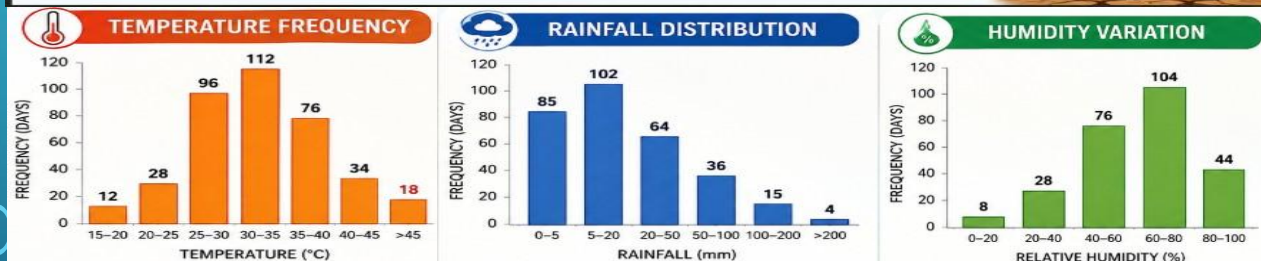
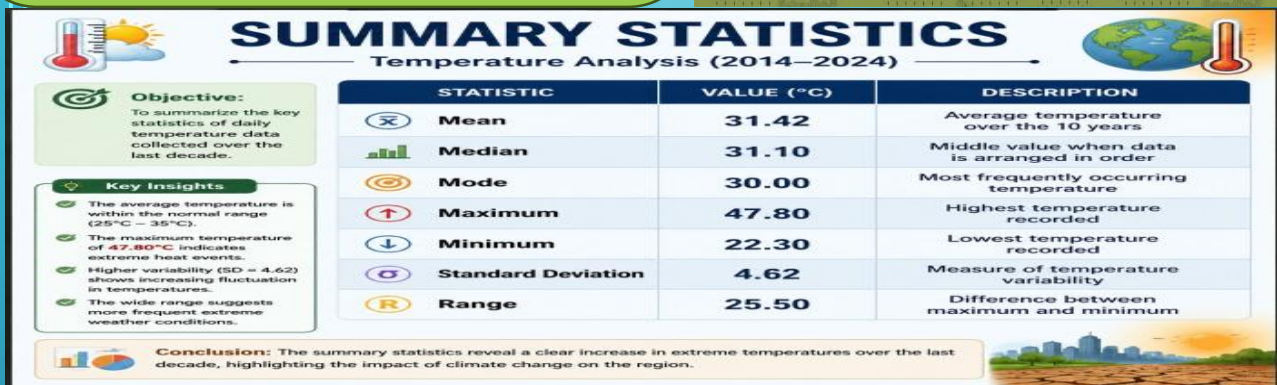
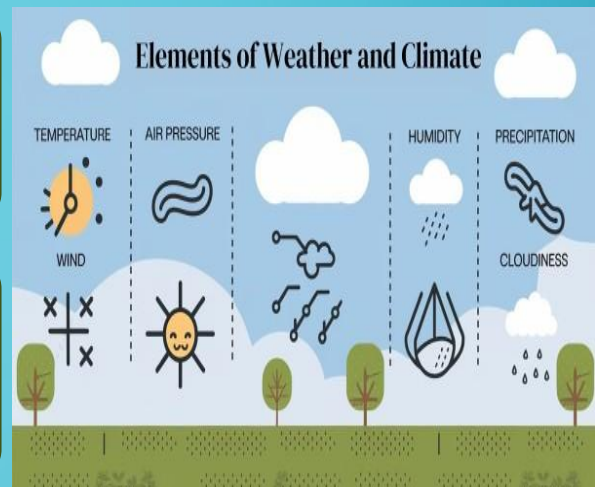
Climate Change and Weather Analysis

Introduction :

- Normal temperature: 25°C–35°C
- Recent decade: Several days above 45°C
- Objective: Analyse climate data using statistical methods.

Distribution of Climatic Variables:

- Temperature Frequency
- Rainfall Distribution
- Humidity Variation



Conclusion :

Statistical analysis clearly shows an increase in extreme temperature events, supporting evidence of climate change and highlighting the need for immediate policy action.



SOCIAL MEDIA ENGAGEMENT ANALYSIS

ANALYZING 500 VIDEOS UPLOADED BY A CONTENT CREATOR

Most videos receive fewer than 10,000 views, while a few viral videos receive over 10 million views.



1 DATA SUMMARIZATION METHODS

Summary Statistics - Views

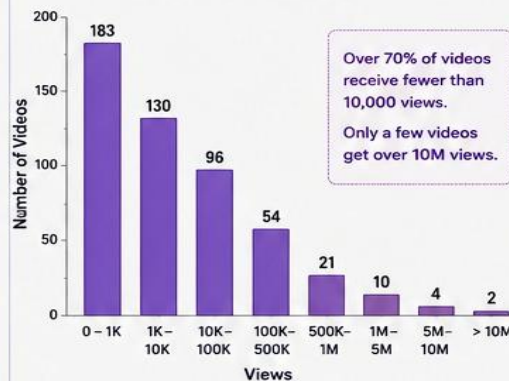
Total Videos	500
Total Views	1,252,743,820
Mean Views	2,505,488
Median Views	7,842
Mode	3,215
Minimum Views	532
Maximum Views	28,400,000
Standard Deviation	3,842,335
Q1 (25th Percentile)	2,150
Q3 (75th Percentile)	21,300
IQR	19,150

Key Takeaways

- ✓ Mean is much higher than median, indicating right-skewed data.
- ✓ Most videos have low views, but a few get extremely high views.
- ✓ High standard deviation shows large variability in engagement.

2 DISTRIBUTION OF ENGAGEMENT METRICS

Distribution of Video Views

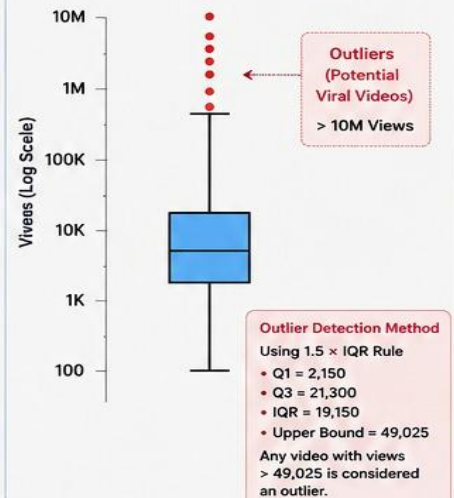


Over 70% of videos receive fewer than 10,000 views. Only a few videos get over 10M views.

<10K Views 313 videos (62.6%)	10K - 1M Views 166 videos (33.2%)	>1M Views 21 videos (4.2%)
----------------------------------	--------------------------------------	-------------------------------

3 DETECTION OF VIRAL-CONTENT OUTLIERS

Box Plot of Video Views (Log Scale)



4 SHOULD VIRAL VIDEOS BE RETAINED?

Why Retain Viral Videos?

- ✓ Drive massive reach & awareness
- ✓ Boost followers & subscriptions
- ✓ Improve overall channel credibility
- ✓ Increase ad revenue & brand deals
- ✓ Help older content get discovered

Potential Concerns

- ✗ Skewed metrics may mislead analysis
- ✗ Unrealistic performance expectations
- ✗ May not reflect consistent engagement

Conclusion: YES

Retain viral videos, but analyze both with and without outliers for better insights.

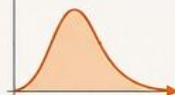


5 INTERPRETATION OF SKEWNESS AND KURTOSIS

Skewness 5.68

Positive Skewness

A long tail on the right indicates a few videos get extremely high views.



Kurtosis 42.31

High Kurtosis (Leptokurtic)

Distribution has heavy tails and a sharper peak, meaning more extreme outliers than normal distribution.



Interpretation

Engagement is highly concentrated: Most videos get low views, while few go viral and create huge impact.



6 BUSINESS IMPLICATIONS FOR CONTENT CREATORS



Performance Expectations

Understand that viral success is rare. Focus on consistent content creation.



Revenue Planning

High variability in views affects ad revenue. Diversify income streams.



Content Strategy

Identify patterns in viral videos (topic, format, length, timing).



Audience Growth

Viral videos drive massive audience growth and brand recognition.



Resource Allocation

Invest in high-potential content areas & engagement tactics.

7 STRATEGIES TO IMPROVE AUDIENCE ENGAGEMENT



Know Your Audience
Understand audience interests, age, location and watch behavior.



Create High-Quality Content
Focus on value, storytelling, and visual quality.



Optimize Titles & Thumbnails
Eye-catching thumbnails and compelling titles increase CTR.



Post Consistently
Maintain a regular upload schedule to build audience trust.



Engage With Audience
Reply to comments, ask questions, and build community.



Analyze & Adapt
Use analytics to track performance and improve continuously.



Leverage Trends
Jump on trending topics, challenges, and formats relevant to your niche.



KEY INSIGHT

In the world of content creation, consistency brings results, but creativity creates virality!

“

Viral videos are the fuel, but consistent content is the engine that drives long-term success.



FINAL TAKEAWAY

Focus on creating value consistently, learn from data, and keep experimenting to maximize engagement and growth.



STOCK MARKET VOLATILITY ANALYSIS



A stock usually fluctuates within $\pm 2\%$ daily.

Following a major corporate announcement, the stock price rises by **30%** in a single day.

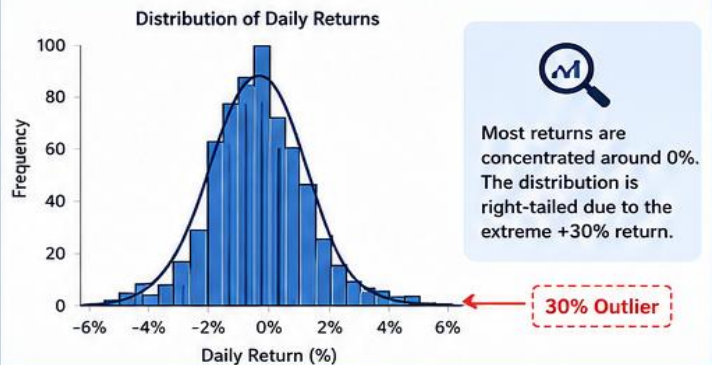
1 SUMMARY STATISTICS OF STOCK RETURNS

Metric	Value
Mean Daily Return	0.15%
Median Daily Return	0.12%
Standard Deviation	1.85%
Minimum Return	-4.60%
Maximum Return	30.00%
Skewness	2.45
Kurtosis	8.67
Observations	252 Days

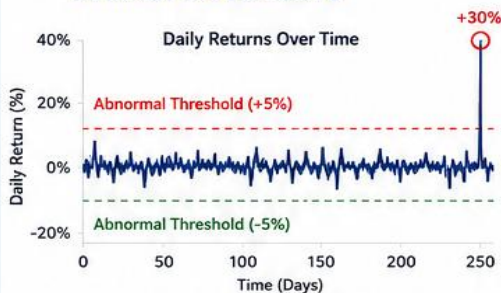


The average daily return is slightly positive. High standard deviation indicates volatility. The 30% spike is far above the usual range ($\pm 2\%$).

2 DISTRIBUTION OF FINANCIAL DATA



3 DETECTION OF ABNORMAL MARKET MOVEMENTS



4 TREATMENT OF EXTREME OBSERVATIONS



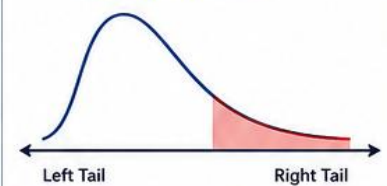
Extreme values can distort analysis and risk estimation. Treatment methods:

- Winsorization (e.g., cap at 5th and 95th percentile)
- Trimming (remove extreme outliers)
- Robust statistical methods



For this analysis, Winsorization is applied to reduce the impact of the +30% outlier while retaining important information.

5 INTERPRETATION OF SKEWNESS

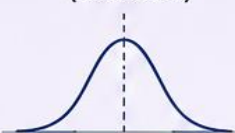


Skewness = +2.45 (Positive Skew)

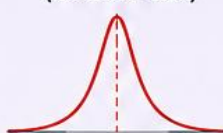
The distribution has a long right tail, indicating the presence of large positive returns (e.g., +30% event). This means extreme gains are more likely than extreme losses.

6 IMPORTANCE OF KURTOSIS IN FINANCIAL RISK ANALYSIS

Normal Distribution
(Kurtosis ≈ 3)



High Kurtosis
(Kurtosis = 8.67)



Kurtosis measures the "tailedness" of the distribution.



High kurtosis (>3) indicates fat tails and a higher chance of extreme events.



Our value (8.67) is much higher than 3 $\rightarrow$ significant risk of extreme market movements.



Investors should be cautious as large gains or losses are more probable than in a normal market.

7 INVESTMENT RECOMMENDATIONS BASED ON FINDINGS

1. Stay Informed



Corporate announcements can cause sharp price movements. Stay updated.

2. Manage Risk



Use stop-loss orders and proper position sizing to limit downside risk.

3. Diversify



Diversification reduces the impact of extreme movements in a single stock.

4. Long-Term View



Ignore short-term spikes and focus on fundamental strength and long-term growth.

5. Monitor Volatility



High volatility periods require active monitoring and risk adjustment.

KEY TAKEAWAY

The +30% jump is an abnormal event with high impact. Through statistical analysis, we understand the behavior of returns, identify risks, and make informed investment decisions.



Data-driven insights lead to smarter and safer investments.

“In the short run, the market is a voting machine; in the long run, it is a weighing machine.” – Benjamin Graham



SMART CITY TRAFFIC MANAGEMENT SYSTEM

A smart city uses sensors to monitor traffic flow.

Average traffic volume is **700** vehicles per hour.
During a major cultural festival, traffic increases
to **7,000** vehicles per hour.

1 DATA SUMMARIZATION OF TRAFFIC PATTERNS

Summary Statistics (Vehicles per Hour)

Metric	Normal Days	Festival Day
Mean	700	7,000
Median	680	6,850
Minimum	320	3,200
Maximum	1,150	9,800
Standard Deviation	210	1,420
Variance	44,100	2,016,400
25th Percentile (Q1)	560	5,900
75th Percentile (Q3)	820	7,900
Observations	720 hours	24 hours*

*Festival data collected continuously for 24 hours.



Average traffic during normal days is **700** vehicles/hr.



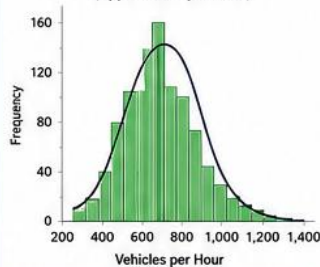
During festival, traffic increases **10 times** to **7,000** vehicles/hr.



Maximum traffic spikes to **9,800** vehicles/hr.

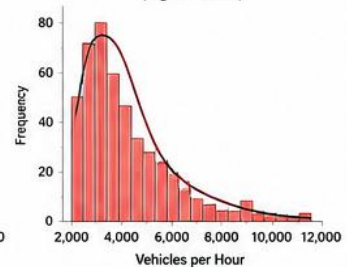
2 ANALYSIS OF TRAFFIC DISTRIBUTION

Normal Day Traffic Distribution
(Approximately Normal)



Most traffic volumes are concentrated around the mean (700 vehicles/hr) with a bell-shaped distribution.

Festival Day Traffic Distribution
(Right-Skewed)



Distribution is right-skewed with a long tail due to extremely high traffic volumes.

3 IDENTIFICATION OF UNUSUAL TRAFFIC EVENTS

Traffic Volume Over Time (Festival Day)

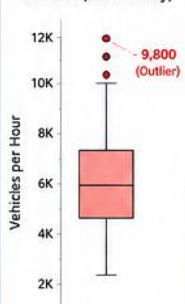


Unusual Event!
Traffic > 2,000 vehicles/hr

Traffic volume crossing 2,000 vehicles/hr is considered unusual compared to normal patterns.

4 OUTLIER DETECTION & TREATMENT

Box Plot (Festival Day)



Outlier Detection

Using IQR Method:
Upper Bound = $Q3 + 1.5 \times IQR$
 $= 7,900 + 1.5 \times 2,000$
 $= 10,900$
Values > 10,900 are considered outliers.

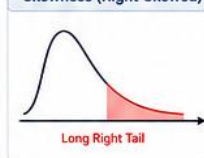
Treatment Methods

- ✓ Capping (Winsorization)
- ✓ Trim extreme values
- ✓ Transformation (log)
- ✓ Robust statistical methods

Outliers are handled to avoid distortion in analysis and modeling.

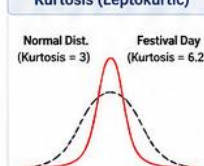
5 ANALYSIS OF SKEWNESS & KURTOSIS

Skewness (Right-Skewed)



Normal Day: Skewness = 0.35 (Nearly Symmetric)
Festival Day: Skewness = 1.85 (Positively Skewed)
High positive skewness indicates that most values are below the mean, with few extremely high traffic volumes.

Kurtosis (Leptokurtic)



High kurtosis (>3) indicates heavy tails and more extreme events compared to a normal distribution.

Implication: Higher probability of extreme traffic congestion.

6 TRAFFIC MANAGEMENT RECOMMENDATIONS



Dynamic Signal Control

Adjust signal timing in real-time based on live traffic density from sensors.



Alternate Route Guidance

Provide real-time route suggestions via mobile apps and variable message signs.



Public Transport Boost

Increase frequency of buses/metro during events to reduce private vehicle usage.



Smart Parking Management

Guide drivers to available parking spaces to reduce search traffic and congestion.



Event-Based Planning

Use historical data to plan resources, traffic police deployment and infrastructure support.



Smart Data. Smarter Decisions. Smoother Cities.

7 PREDICTIVE MODEL FOR FUTURE EVENTS

Model Approach



Use historical traffic data (normal days + past events).



Features: Time of day, Day type, Weather, Event type, Holidays.

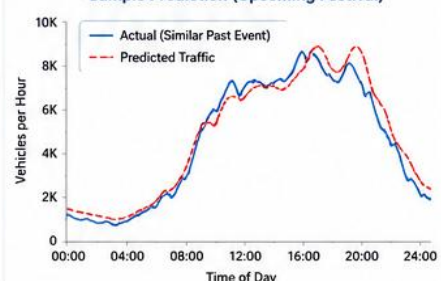


Model: SARIMA / Random Forest / Gradient Boosting.



Predict traffic volume hourly for upcoming events.

Sample Prediction (Upcoming Festival)



Benefits

- ✓ Better resource allocation
- ✓ Reduced congestion
- ✓ Improved citizen experience
- ✓ Lower fuel consumption
- ✓ Reduced pollution
- ✓ Smarter cities



By using data analytics, statistical analysis and predictive modeling, smart cities can efficiently manage traffic, handle unusual events and ensure smooth mobility for citizens.